4.4 Volume of a prism

A box of chocolates is the shape of a **triangular prism**.

The triangular ends each have an area of 140 cm^2.

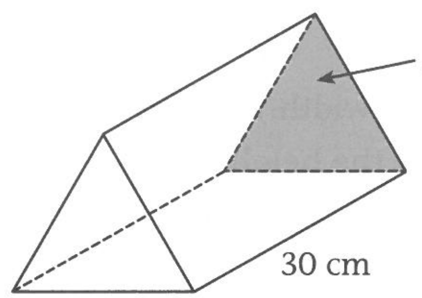
The box is 30 cm long.

The **volume of a prism** can be calculated using the rule:

Volume = area of face × height

$V = A \times h$

EXAMPLE



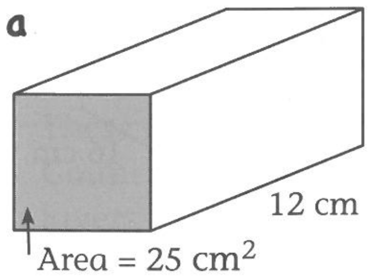
Volume

$= 140 \times 30$

$= 4\,200 \text{ cm}^3$

1 Use the rule $V = A \times h$ to calculate the volume of each prism.

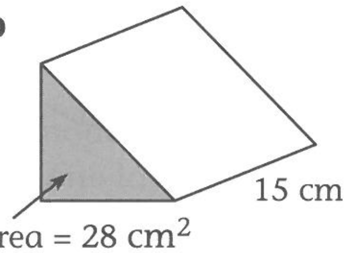
a



Volume = __________

= __________

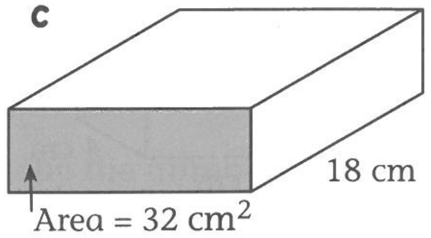
b



Volume = __________

= __________

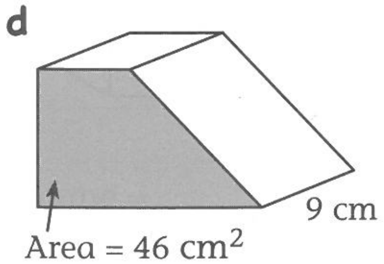
c



Volume = __________

= __________

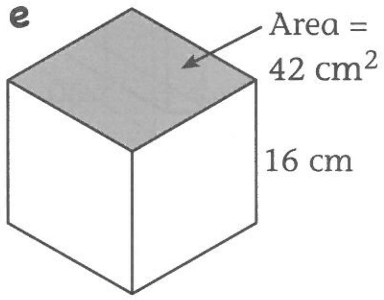
d



Volume = __________

= __________

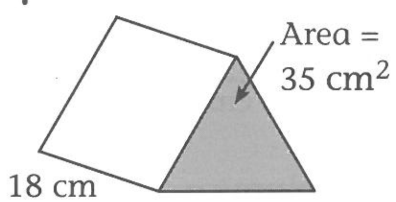
e



Volume = __________

= __________

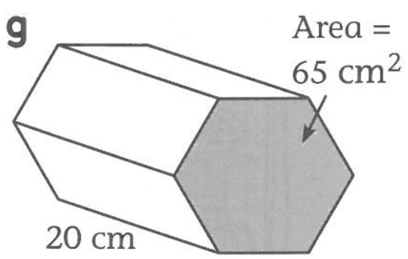
f



Volume = __________

= __________

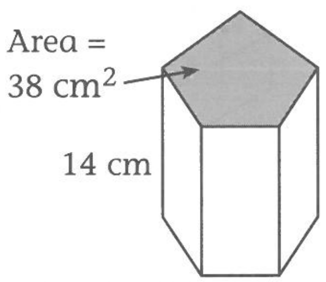
g



Volume = __________

= __________

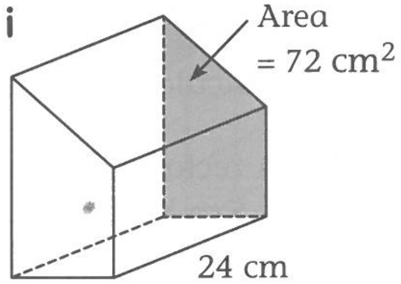
h



Volume = __________

= __________

i



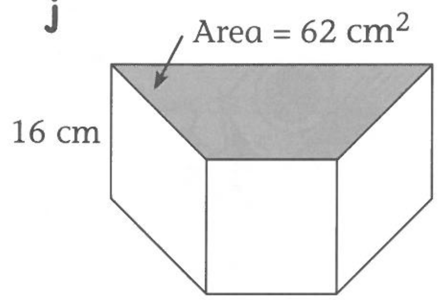
Volume = __________

= __________

4.4 Volume of a prism

CONTINUED

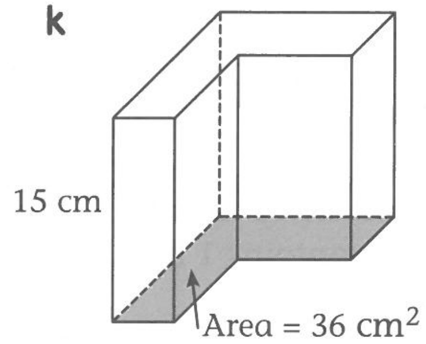
j



Volume = __________

= __________

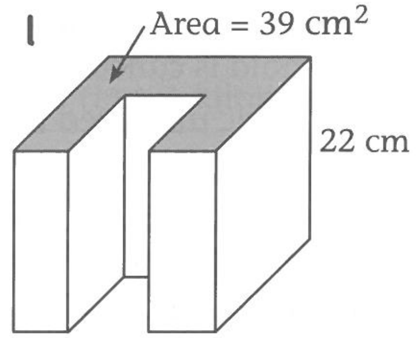
k



Volume = __________

= __________

l

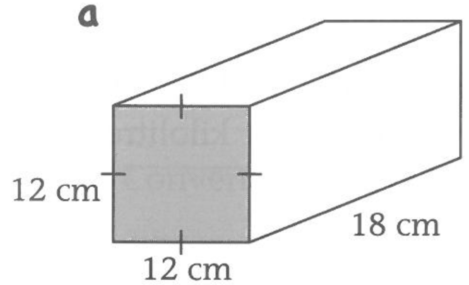


Volume = __________

= __________

2 For each prism, first calculate the area of the cross-section (the shaded face). Then calculate the volume of the prism.

a



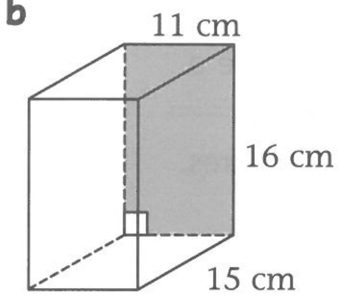
Area = __________

= __________

Volume = __________

= __________

b



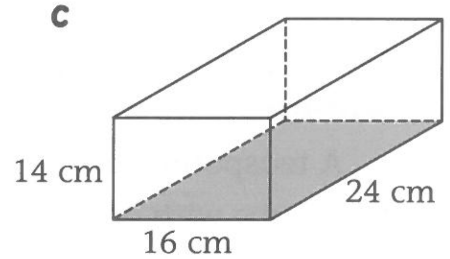
Area = __________

= __________

Volume = __________

= __________

c



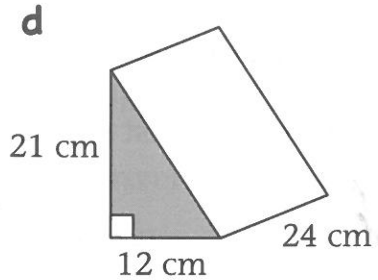
Area = __________

= __________

Volume = __________

= __________

d



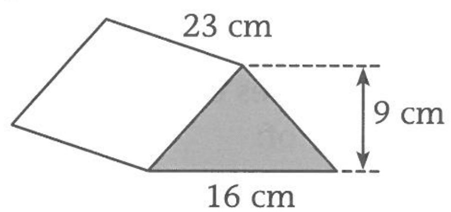
Area = __________

= __________

Volume = __________

= __________

e



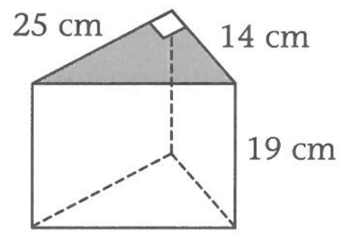
Area = __________

= __________

Volume = __________

= __________

f



Area = __________

= __________

Volume = __________

= __________